GEOGRAPHIC INFORMATION ANALYSIS
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Chapter 8

Local Statistics

CHAPTER OBJECTIVES

In this chapter, we:

- Explain the concepts underlying the emerging array of *local statistics*
- Account for the relatively late arrival of local statistics on the spatial analytic scene
- Review the various approaches that can be used to construct *localities* for the development of local statistics
- Discuss how the popular Getis-Ord family of G statistics are calculated and interpreted
- Outline the local version of Moran's I statistic
- Explain why inference based on local statistics is challenging and describe current approaches to dealing with the difficulties
- Provide an overview of the increasingly popular method *geographically weighted regression*
- Explain how many other spatial analysis methods can be considered as local statistics even if this was not the intent behind their original development

After reading this chapter, you should be able to:

- Explain what is meant by local statistics and suggest reasons for their current popularity
- Provide some explanations for the slow adoption of local statistical approaches
- Review a number of bases on which localities for local statistics can be constructed

- Define the Getis-Ord G and local Moran's I statistics and discuss how they should be interpreted
- Give an account of why inference for local statistics is difficult and outline approaches for addressing the various problems
- In general terms, describe how geographically weighted regression works
- Redescribe a selection of other spatial analytical methods as local statistics

8.1. INTRODUCTION: THINK GEOGRAPHICALLY, MEASURE LOCALLY

We now consider one of the most important innovations in geographic information analysis in recent years, namely, the development and use of a variety of *local statistics*. As we will see, local statistics arise naturally out of any of the methods for measuring spatial autocorrelation, discussed in the previous chapter. Once the connection is noted and generalized, the way is open to development of localized variants of almost any standard summary statistic, with a particularly interesting recent innovation being *geographically weighted regression*. In the same way, many older spatial analysis methods can also be reinterpreted as local statistics. Thus, this chapter is also a useful precursor to the interpolation methods discussed in more detail in the next chapter, because many estimates produced by spatial interpolation are based on local statistics.

What do we mean by a local statistic? A *local statistic* is any descriptive statistic associated with a spatial data set whose value varies from place to place. In the broadest sense, any spatial data set is a collection of local statistics, in that the recorded attribute values are different at each location. A local statistic is different in that it is derived by considering a subset of the spatial data *local* to the spatial location where it is being calculated. A simple example is a localized mean, calculated by determining the mean value of an attribute based on attribute values in the data set near the location of interest. In the next chapter, we will see that such a localized mean is the underlying basis for many simple methods of spatial interpolation. It is also worth noting that a localized mean is exactly equivalent to one kind of smoothing filter that may be applied to image or raster data. In Section 9.5, we will see that this concept has been generalized as *map algebra*, specifically in the form of focal operations on raster data. Thus, the concept of a local statistic is widely deployed in spatial analysis, although it goes by different names in different contexts. For now, the important point is to realize how central the concept of a local statistic is for many spatial analysis methods.

It is perhaps surprising that such a key idea has only gained currency since the mid-1990s. Two review papers in *Progress in Human Geography* by Unwin (1996) and Fotheringham (1997) were the first to highlight explicitly the importance of local statistics. With this in mind, it is useful to consider why the idea has only taken off recently. One important consideration is the mapping capability provided by GIS tools. As we shall see, many local statistics are a natural by-product of the calculation of summary global statistics. Prior to the possibility of easily mapping any data set, the summary result would have been reported and the local statistics used in its calculation discarded. It was the advent of readily available mapping tools that led to the exploration of the potential of local statistics as an analytical output in their own right. These developments parallel the increasing importance of exploratory data analysis (Tukey, 1977), an approach where indentifying outliers and the overall structure in data are important aims and visualization methods are central.

A second technical reason for the recent increase in the popularity of local statistics is that (perhaps paradoxically) the statistical evaluation of local statistics is more challenging than the statistical assessment of related global measures. This parallels the difficulties faced in identifying local clusters in point patterns relative to simply determining if a point pattern is clustered or not. The difficulty lies in assessing analytically the statistical significance of particular localized patterns, and in this context, Monte Carlo simulation approaches are often used to generate pseudo-significance results. The computational burden of simulation-based methods is significant, so local statistics have only become practical as substantial computing power has become generally available.

A third reason for the increased interest in local statistics is recognition of the importance of geographic variation in phenomena. This is itself a side-effect of the widespread adoption and use of GIS tools and the accompanying increase in data availability. As more data have become available, this has allowed studies both to expand their spatial range and to focus in at higher spatial resolution. Both developments have prompted the realization that the idea of a single global process or model being a realistic explanation is not always very plausible.

Finally, interest in local statistics reflects developments in the spatial sciences more generally that increasingly recognize the importance of local contexts in understanding the global patterns of phenomena. While the methods deployed by many human geographers in their research have become increasingly qualitative since the 1980s, this is largely a response to the realization that local contexts matter. Qualitative methods such as interviews and focus groups recognize the multilayered richness of local contexts and the importance of multiple interpretations by people in those contexts. By their very nature, quantitative or statistical data tend to simplify and flatten out some of that complexity. This is particularly the case if we restrict ourselves to global summary statistics and a search for broad generalizations about the whole of a large study area. Local statistics, on the other hand, emphasize the variety among local contexts and focus attention on what is different from one place to another rather than on what is similar. Thus, local statistics can be seen as a response from the quantitative end of geography to increased recognition of the importance of local context.

8.2. DEFINING THE LOCAL: SPATIAL STRUCTURE (AGAIN)

In Section 7.4, as a necessary precursor to the development of global spatial autocorrelation statistics, the construction of a wide variety of spatial weights matrices among a set of polygons was described. The local neighborhood of a particular location is fully described by a single row in such spatial weights matrix. Thus, if the weights matrix is

$$\mathbf{W} = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & & \vdots \\ \vdots & & \ddots & \vdots \\ w_{n1} & \cdots & \cdots & w_{nn} \end{bmatrix} \tag{8.1}$$

then a row matrix

$$\mathbf{W}_i = [w_{i1} \quad w_{i2} \quad \cdots \quad w_{in}] \tag{8.2}$$

describes the local neighborhood of each location i. As has been discussed in Chapters 2 and 7, there is a wide range of possible bases on which the weights matrix and thus localities may be defined.

Some More Revision

It is useful at this point to revisit Section 2.3 and revise the materials on definitions of distance, adjacency, neighborhood, and proximity, as well as how these can be summarized in adjacency A or weights matrices **W**.

When you have done this, revisit Section 7.4 to remind yourself of how these matrices are used in the definition of global spatial autocorrelation indices.

For polygon data, adjacency either immediately or indirectly via intervening neighboring polygons is a common basis for construction of localities. Using polygon centroids, localities may be constructed based on distance criteria, and the same approach can also be applied to point data sets. In this case, it is also possible to introduce additional constraints so that all locations have some minimum number of neighbors in their locality.

It is important to keep in mind that the choices made in constructing localities prior to determining local statistics are a critical aspect of the analysis. Local statistics may point to patterns of a particular kind when localities are constructed based on adjacency, but they may reveal completely different patterns when localities are constructed based on a distance criterion. The important points are, first, that where possible, a number of different weights matrix constructions be examined, and second, that consideration be given to which method makes the most sense in substantive terms. For example, it is easy to assume that simple spatial adjacency based on contiguity among a set of polygons is somehow the "natural" approach to constructing localities. However, when we are interested in some phenomenon whose patterns are likely to be related to transport accessibility, it may be much more relevant to connect locations via the transport network, and thus to base adjacency on estimated distances over road or other networks. Such options have become much more readily explored using the capacity of GIS to relate spatial data in a wide range of ways.

8.3. AN EXAMPLE: THE GETIS-ORD G_i AND G_i^* STATISTICS

The goal of the Getis-Ord family of local statistics developed in Getis and Ord (1992) and Ord and Getis (1995) is to enable detection of local concentrations of high or low values in an attribute, and it nicely illustrates the concept of a local statistic. The statistic is simple to calculate. For a location i, the value is given by

$$G_i(d) = \frac{\sum_j w_{ij}(d) x_j}{\sum_{j=1}^{n} x_j} \quad \text{for all } i \neq j \tag{8.3}$$

where $w_{ij}(d)$ are weights from the spatial weights matrix and x_j denotes the attribute values at locations j. The dependence on a particular set of assumptions about spatial dependence is denoted for both G_i and w_{ij} by their functional dependence on d. Note that the numerator in this fraction is the sum of the x_j values in the locality of the location of interest i but not including x_i itself, and that the denominator is the sum of all the x values in the whole study area. Thus, G_i is simply the proportion of the sum of all x values in the study area accounted for by just the neighbors of i. In a location where high values are clustered, G_i will be relatively high; conversely, in a location where low values are concentrated, G_i will be low. The closely related statistic G_i^* is defined similarly to G_i, the only difference being that the attribute value at location i itself is included in both the numerator and denominator summations in Equation (8.3). Due to the dependence of G_i (and G_i^*) on the ratio of two sums of x values, it is important that the attribute under consideration be a ratio-scaled variable with a natural origin. Another way to think about this is that the value of G_i will be different if we add a constant value to every location or if we transform the variable by taking logarithms.

It is relatively easy to derive expected values and the variance for the G_i statistic under an assumption of random spatial distribution of the attribute values. The expected value is given by

$$E(G_i(d)) = \frac{\sum_j w_{ij}(d)}{n-1} \tag{8.4}$$

which states that the expected value of G_i is that proportion of the study region accounted for by the neighborhood of location i where we assume 0/1 valued adjacency weights. Calculation of the statistic's variance is more complex (see Getis and Ord, 1992, p. 191 for details).

In all the equations above, d denotes the fact that we can calculate the value of the statistic for various distances or, more generally, under a variety of definitions of locality. Thus, as discussed in the previous section, the spatial weights matrix is chosen by the analyst under some assumption of what constitutes a meaningful set of localities for the purposes of the analysis. In essence, the choice of a $\mathbf{W}$ encapsulates a hypothesis about both the range and nature of any likely local geographic effects. Typically, this will be distance-dependent in some way, insofar as we are interested in knowing if concentrations of high or low data values occur only at short distances, over a wide range of distances, or perhaps only at large distances.

Because expected values and variances for the Getis-Ord statistic are known, a z score can be determined for each location's G_i value. The map in Figure 8.1 shows calculated G_i values as z scores based on the tuberculosis incidence data considered in the previous chapter, using the Rook's case adjacency weights matrix. The notable difference between this map and the map of incidence rates themselves shown in Figure 7.7 is that the

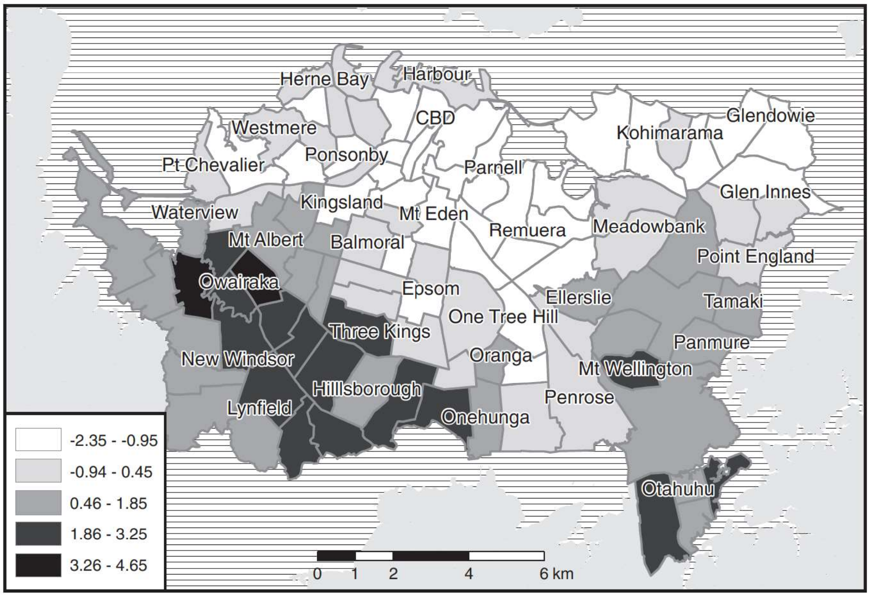

Figure 8.1 Map of the Auckland tuberculosis data z scores determined from the calculated G_i values.

highest-incidence locations are not the ones with the highest associated G_i values. Instead, the census area units neighboring high-incidence areas are highlighted. This is most obvious in the case of Mt. Wellington, toward the southeast of the area, which has a high z score in spite of having a relatively low incidence rate. The two particularly high Getis-Ord statistic scores in the western part of the map are also not locations among the highest incidence rates in the original map, but they do have neighboring locations with high incidence rates.

Although we might normally interpret z scores outside the range -1.96 to $+1.96$ as unusual cases and single out these parts of the map for particular attention, more care is required in making inferences from local statistics. This is because an assumption of normality in the distribution of most local statistics is problematic, particularly where the localities under consideration are small, so that the statistic is being calculated based on small numbers of cases. If d is increased so that localities have larger numbers of cases, this is less of a problem, but, of course, the localities under consideration are no longer quite so local either! It is particularly important to be cautious about overinterpreting high or low z score values toward the edges of the study area, as these may be based on very small numbers of locations. These considerations limit the usefulness of analytical approaches to the statistical assessment of the statistic, leading to a need for simulation-based approaches to inference. This difficulty is common in the interpretation and analysis of local statistics and is considered in more detail in Section 8.4.

Other Local Statistics

In principle, almost any standard statistic can be turned into a local statistic. Instead of summarizing over a whole data set, we summarize over only the data in the locality of each data point. The calculations required to determine a global value of Moran's I provide another example. In this case, at each location, the following quantity is calculated:

$$I_i = z_i \sum_j w_{ij} z_j \tag{8.5}$$

where the z values are z scores determined from the values of the attribute of interest for the whole data set. Positive values of I_i result where either low or high values of the attribute are near one another, while negative values result where low and high values are found in the same area of the map. Thus, local Moran's I gives an indication of data homogeneity and diversity. This statistic is fully developed in a paper by Luc Anselin (1995), which presents the more general concept of *local indicators of spatial association* (LISA) statistics.

When working with the local version of Moran's I, the *Moran scatterplot* of Figure 7.8 comes into its own as an analytical tool. The four quadrants of the plot defined by the global mean attribute value (or by $z_i = 0$ and $\sum_j w_{ij} z_j = 0$) each correspond to the different possible combinations of the value at i and among its neighbors. We can identify these in shorthand as "low–low" or "high–high" cases that contribute to positive autocorrelation and "low–high" or "high–low" cases that contribute to negative autocorrelation. For example, "high–high" cases are ones where the value of i is high and neighboring values are also high. While all cases are in one of the four quadrants, in general, we are only interested in cases that are statistically unusual in some sense. How these cases are identified is discussed in Section 8.4 when we consider inference for local statistics.

Getis and Ord (1992, pp. 198–199) suggest that both the G_i and I_i statistics should be used in any exploration of a spatial data set, as they measure different things and may point to different driving processes underlying the observed spatial distributions of attribute values. While Moran's global statistic measures spatial autocorrelation without distinguishing between patterns dominated by concentrations of high or low values, a global version of the G_i statistics enables these cases to be distinguished. This is clearer when the global G statistic is written out in full:

$$G(d) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(d)x_i x_j}{\sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j} \quad \text{for all } i \neq j \tag{8.6}$$

This statistic tends to have a high value when the locations where high values are located near one another outweigh the locations where low values are located near one another (and vice versa). Thus, G helps to determine whether it is clusters of high values ("hot spots") or low values ("cold spots") that contribute most to an overall finding of positive spatial autocorrelation.

While a local form of almost any statistic could be developed in principle, in practice relatively few have been formalized as local statistics per se, although, as we shall see, many statistics can be usefully considered as local statistics even if they are not used in the exploratory manner discussed here. The likely reason for this is that the most pertinent feature of any spatial data set is the degree to which attribute values exhibit spatial dependence, and this is precisely the aspect of the data on which the Getis-Ord and Moran's statistics focus. Another reason is that local statistics are most useful as exploratory tools in the early stages of an investigation. More formal statistical analysis calls for some method by which significance can be determined, and this presents difficult problems in the context of the small number of cases included in each localized calculation.

8.4. INFERENCE WITH LOCAL STATISTICS

We have seen how the simplifying assumption of spatial randomness can allow analytical results for the expected values of the G_i local statistics to be determined. Such simplifying assumptions treat local statistics as simple random samples from the total population of all the attribute values in the study area. In many cases, because of the central limit theorem, this results in the expectation that local statistics will be normally distributed, and that unusual cases can be identified where calculated z values are less than -1.96 or greater than $+1.96$, which is the range of values associated with standard 95% confidence intervals.

However, this approach is problematic. It is evident that this is the case when Figure 8.1 is reviewed. Six census areas have z score values less than -1.96, while 14 have values greater than 1.96. On a conventional interpretation, this would suggest that almost 20% of the locations (20 out of 103 census area units) are unusual in a statistical sense. The difficulty here is that the data are evidently not well accounted for by a null model that assumes complete spatial randomness. We already know this from the calculations detailed in Section 7.5, which show significant positive spatial autocorrelation in this data set. If we know that the data are positively autocorrelated, it makes little sense to identify statistically unusual cases based on a null model that assumes complete spatial randomness!

Another difficulty is that this is a situation where repeatedly applying a statistical test to the same data, when that test assumes independence of the observations, leads to problems (as was mentioned in relation to the GAM in Section 6.7). Consider any two locations, A and B, that are neighbors in a spatial data set. If A has an unusually high value of the G_i statistic, then, given that it shares many of the same neighbors as B, it is highly likely that B will also have a high value of the G_i statistic. Thus, statistical tests of local statistics are inherently nonindependent, and we must make some adjustment to the criteria we use to determine which observations are unusually high or low. This is known as the *multiple testing problem*, which can be addressed by adjusting the probability threshold used to determine which results are considered statistically significant. Where n tests are conducted, with a desired statistical significance (i.e., a p-value) of α, one possible corrected significance level suggested by Sidak (1967) and endorsed by Ord and Getis (1995) and by Anselin (1995) is

$$a' = 1 - (1 - a)^{1/n} \tag{8.7}$$

An alternative simpler approach, known as the *Bonferroni correction*, is to set $a' = a/n$. In practice, the two approaches produce very similar corrected significance levels. In Figure 8.1, where $n = 103$, for a 0.05 significance level, applying the correction of Equation (8.7) gives an adjusted p-value of 0.000498, while the Bonferroni correction produces a similar value of 0.000485. The former value is associated with a z score of ± 3.29.

Applying this new threshold to the mapped data results in only the two highest-value census area units (those located east and west of Owairaka) being considered statistically significant cases. We would interpret this as meaning that even given the known positive autocorrelation in these data, those locations exhibit unusually high similarity to their neighbors. Some writers consider these corrections for multiple testing to be too conservative for the *semi-independent* tests actually applied in the context of local statistics, and also believe that they result in too few determinations of statistically unusual cases. Anselin (1995, p. 96) discusses this matter in some detail. The crux of the argument is that the standard corrections suggested above are for cases where *exactly the same data* are tested n times. For local statistics, overlapping subsets of the same data are tested a number of times, *but never exactly the same subsets*. A rough estimate of the "effective" number of multiple tests actually occurring might be $\sum_i \sum_j w_{ij}/n$, where n is the number of locations in the data set. However, no detailed research results have been reported in this area.

A more recent (and liberal) approach to the statistical assessment of local statistics is to apply a Monte Carlo simulation procedure to produce pseudosignificance values. This is the same approach adopted in assessing many point pattern measures discussed in Section 5.4. For local statistics, the approach is typically repeated using *conditional permutation* (or "shuffling") of the attribute values in the spatial data among the locations in the data set. Each time the data are shuffled, the value at the location of interest is held constant (this is what makes the permutation conditional), the calculations for the statistic in question are performed on the shuffled data, and the resulting value of the local statistic is determined. The permutation procedure is repeated a large number of times (say, 999), and the value of the local statistic associated with the actual distribution of the attribute is ranked relative to the list of values produced by the permutation procedure. Measured actual values of the local statistic that are either very low or very high relative to the list of results produced by the shuffling procedure are then judged to be of interest. A pseudosignificance value can be determined by noting the rank of the actual local statistic relative to the permuted results. For example, if the actual local statistic is the highest recorded among 999 permutations, then it is estimated to be a 1 in 1000 occurrence with a pseudosignificance of $p \sim 0.001$.

While this approach is more computationally intensive than the results derived from analytical expected values and variances, it is conceptually more satisfying and has become routine given contemporary computational resources on the desktop. With appropriate adjustment of the permutation procedure, the simulation approach also has the potential to be used to explore how unusual are the values of local statistics given the presence of known levels of global spatial autocorrelation. This is an aspect of the results reported by Anselin (1995, pp. 108–111) that deserves more attention, since his results clearly demonstrate that the distributional characteristics of local statistics can be expected to depend strongly on global levels of autocorrelation. An appropriately designed permutation procedure that maintains levels of global autocorrelation in the permuted data sets similar to those observed in the actual data would, at least in theory, enable identification of those local patterns that are unusual even in the context of the observed global patterns. We are not aware of any reported results of this kind, and it is clear that such analysis would present substantial challenges in both execution and interpretation. Given such challenges, it seems likely that local statistics of the kind discussed in this section will continue to be used primarily in an exploratory and descriptive mode for the foreseeable future. Many examples of the use of these measures can be found in the research literature in a wide range of subject areas, in spite of the lack of a completely satisfactory inferential framework. As evidence of this, consider the fact that at the time of writing, the three papers by Anselin (1995) and Getis and Ord (1992, 1995) have been cited a combined total of almost 1000 times!

8.5. OTHER LOCAL STATISTICS

Geographically Weighted Regression

Another popular local statistic developed in the last decade or so is geographically weighted regression. In a simple multivariate regression model, we model the relationship between one *dependent variable* and one or more *independent variables*. The mathematical model underlying multiple regression is

$$\begin{aligned} y_i &= b_0 + b_1x_{i1} + b_2x_{i2} \cdots + b_mx_{im} \cdots + \varepsilon_i \\ &= b_0 + \sum_{j=1}^{m} b_jx_{ij} + \varepsilon_i \end{aligned} \tag{8.8}$$

so that the value of the independent variable at each location y_i is modeled as the sum of a constant b_0, a sum of products of each independent variable value x_{ij}, and a coefficient b_j, along with an error term ε_i. The model is fitted to the observed data using a least squares regression procedure, which ensures that the sum of the squared errors at all locations in the data set is minimized. The mathematics underlying regression is comparatively simple, but beyond the scope of this book, and is covered in numerous introductions to statistics. For the present purpose, it is convenient to express the full regression model in matrix terms as follows:

$$\begin{aligned} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} &= \begin{bmatrix} 1 & x_{11} & \cdots & x_{1m} \\ \vdots & & \ddots & \\ 1 & x_{n1} & & x_{nm} \end{bmatrix} \begin{bmatrix} b_0 \\ \vdots \\ b_m \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \\ \varepsilon_n \end{bmatrix} \\ \mathbf{y} &= \mathbf{Xb} + \mathbf{e} \end{aligned} \tag{8.9}$$

so that $\mathbf{b}$ is a vector containing the estimated regression model coefficients, and $\mathbf{y}$ and $\mathbf{X}$ contain the observed data for the dependent and independent